



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 01_Block_Diagram			
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I2C	Sub SYS	Function	Part Number	I2C Spec.	Budget Time	I2C Slave Address / Write / Read (7-bit mode)	
I2C-0	AP	Cap Touch controller	GT1151	400 Kbps	Yes.	0x5D	Write 0xBA / Read 0xBB
		LCM Gate Drive	RT4801HWSC	400 Kbps	Yes.		
		Magnetic Sensor	AK09918C	400 Kbps	Yes.	0x0C	Write 0x18 / Read 0x19
I2C-1 (3C)	AP Sensor Hub	Ambient Light Sensor	CM03558	400 Kbps	Yes.	0x51	Write 0xA2 / Read 0xA3
		Proximity Sensor					
		Pressure Sensor	BMP280	400 Kbps	Yes.	0x77	Write 0xEE / Read 0xEF
						0x1A	Write 0xA4 / Read 0xA5
I2C-2 (3C)	AP	Rear Camera	EEPROM	400 Kbps		0x59	Write 0xA7 / Read 0xA1
			AF driver	400 Kbps		0x5C	Write 0x19 / Read 0x19
						0x20	Write 0xA4 / Read 0xA8
		Dual cam (B+b) - Main1	EEPROM	400 Kbps		0x59	Write 0xA7 / Read 0xA1
			AF driver	400 Kbps		0x5C	Write 0x19 / Read 0x19
I2C-3	AP	NFC	ST21NFCD	400 Kbps	Yes.	0x08	Write 0x10 / Read 0x11
		Dual cam (B+b) - Main2	EEPROM	400 Kbps		0x18	Write 0x20 / Read 0x21
			EEPROM				
			AF driver				
I2C-4 (3C)	AP	Front Camera	EEPROM	400 Kbps		0x20	Write 0xA7 / Read 0xA8
			AF driver	400 Kbps		0x5C	Write 0xA4 / Read 0xA5
I2C-5	AP	Reserved					
I2C-6	AP SSPM	Reserved					
I2C-7	AP	Audio Smart PA	MT5980	400 Kbps	Yes.	0x34	Write 0xA9 / Read 0xA9
		SW Charger	SW Charger	400 Kbps	Yes.		
		CC Detect	RT1715WSC	400 Kbps	Yes.		
		Flash LED 5V Boost	RT4505	400 Kbps	Yes.		

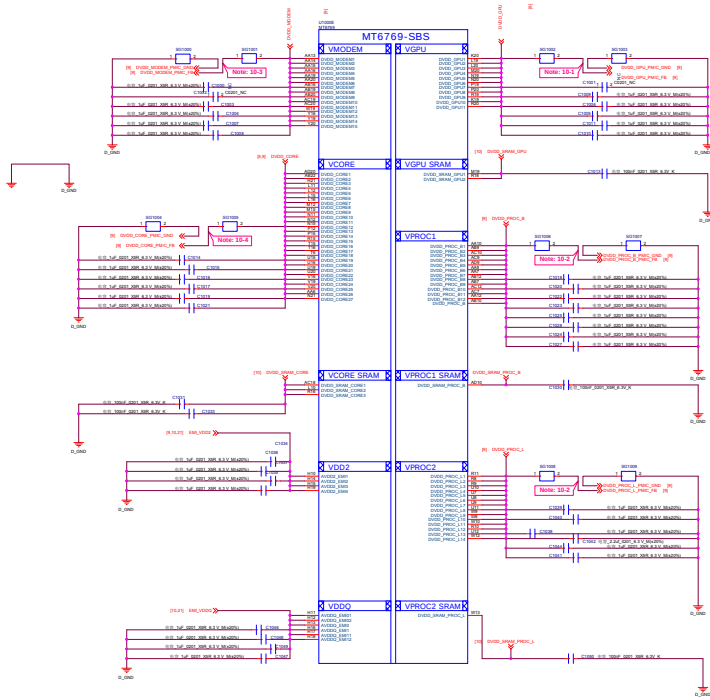
Note : I2C Spec : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

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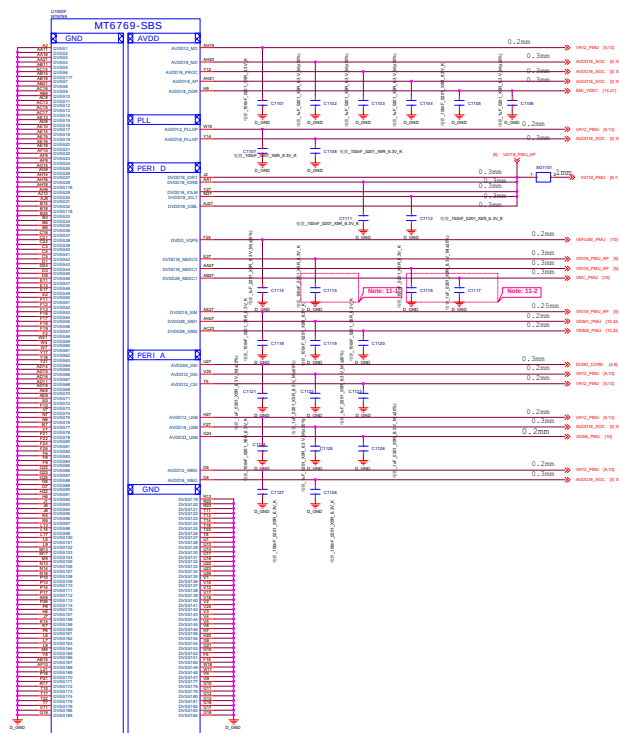
Date	Category	Item
2018.12.14 (V0.1)	Page 04	V0.1 Release
2019.01.11 (V0.2)	Page 12	Add CAM_PDN3(GPIO110) HW pin for Note 12-4: HW pin for DDR type feature in bootstrap Delete R1007 and R1012 ext. pull for LP4X eMCP
	Page 13	Add CAM_PDN3(GPIO110)'s Note : No ext. pull for DDR type feature in bootstrap
	Page 69	Add CAM_PDN3(GPIO110)'s Note : No ext. pull for DDR type feature in bootstrap
	Page 21	Add C2106 for VBIF28 cap
	Page 22	Delete R2202

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Note 10-1:	Differential pair of DVD0_GPU remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)
Note 10-2:	Differential pair of DVD0_PROC remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)
Note 10-3:	Differential pair of DVD0_MODEM remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)
Note 10-4:	Differential pair of DVD0_CORE remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)



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Schematic design notice of "11_BB_POWER_IO" page:

Note 11-1: C1115 closed DVDD18_MSDC0 150mil

Note 11-2: C1116 closed DVDD18_MSDC1 150mil

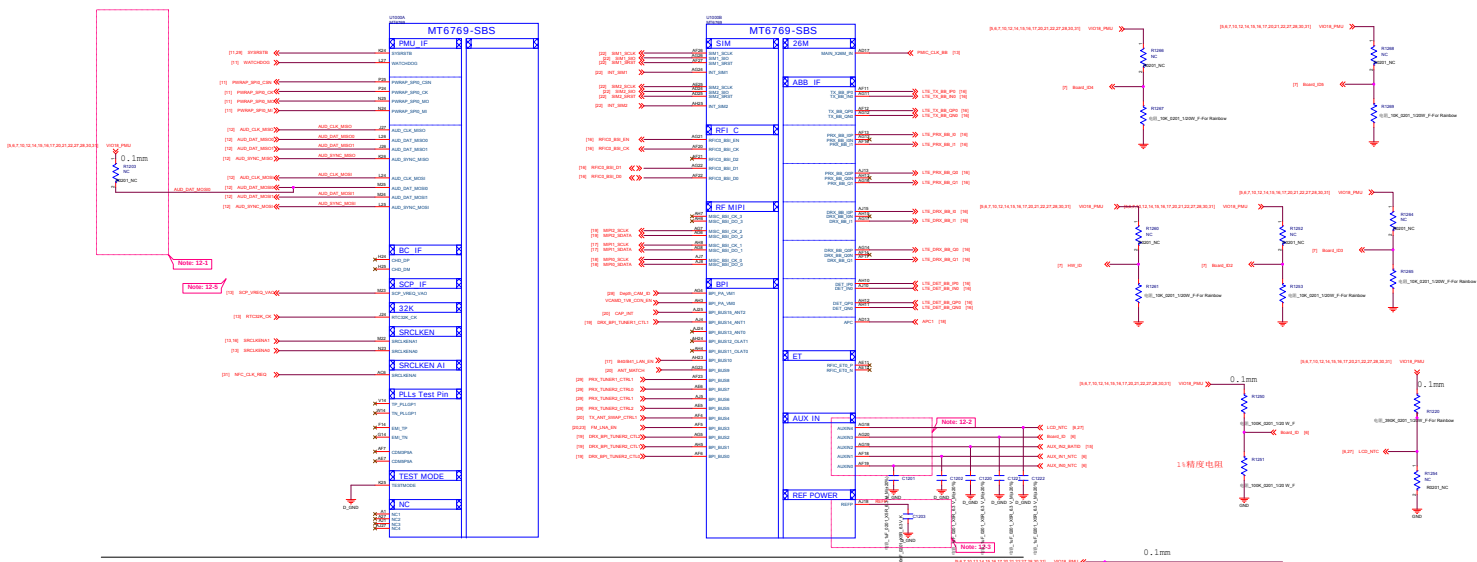
C1117 closed DVDD18_MSDC1 150mil

AVDD12: 单个点走8mil, 两路分支各16mil, 总路径20mil

VDD18_PWD_AP: 总路径40mil

AVDD18_SDC: 总路径20mil

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Schematic design notice of "12_BB_1" page:

Note 12-1: PWRAP_SPI0_CSN and AUD_DAT_MOSI0 are bootstrap pins to select which interface will be the JTAG pin out

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	AP JTAG	JTAG Function
HI	LO	AP JTAG	MD JTAG
LO	HI	SPD + ENTB	SPD + SPD
LO	LO	SPD + ENTB	SPD + SPD
LO	LO	SPD + ENTB	SPD + SPD

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: The de-coupling cap. for REFF (AJ18 ball) have to be placed as close to BB as possible.

Note 12-4: HW pin for DDR type feature in bootstrap (refer to BB HW design Notice)

AUD_SYNC_MISO	AUD_CLK_MISO	CAM_PDN3	PMIC 6555 voltage	DDR Type
default=PD	default=PD	default=PD	VRAM1 / VRAM2	DDR
No ext. pull	No ext. pull	No ext. pull	1.25V / 0.8V	LP4X eMMC
No ext. pull	No ext. pull	No ext. pull	0.8V / 1.0V	LP4X eMMC
12K pull to VDD18	12K pull to VDD18	No ext. pull	1.25V / 0.8V	LP4X eMMC
12K pull to VDD18	12K pull to VDD18	No ext. pull	1.25V / 1.0V	LP4X eMMC

Note 12-5: Charger must have D+/D- pin for charger type USB detection. Charger must have at least 500mA USB current for all charger type.

Thermistor to sense RF PA temperature

Thermistor to sense AP temperature

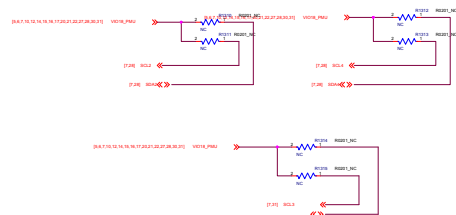
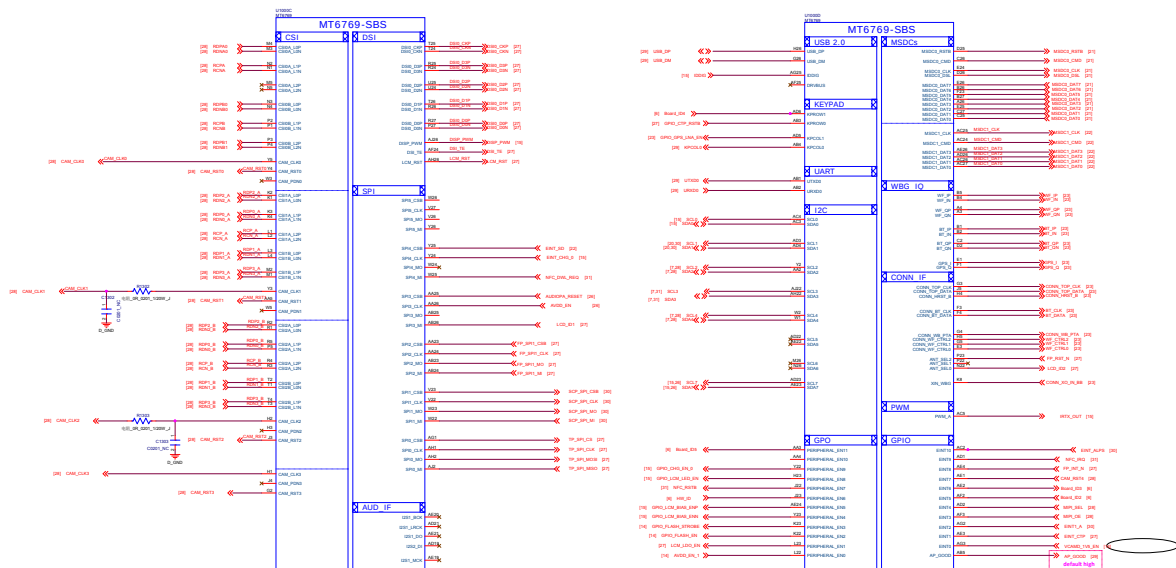
1. B1207 must keep a distance about 0.5mm edge from AP and far from other heat sources to sense AP temp.
2. The distance in the thermal distance from package edge to edge.

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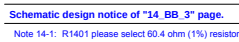
Title 01_Block_Diagram

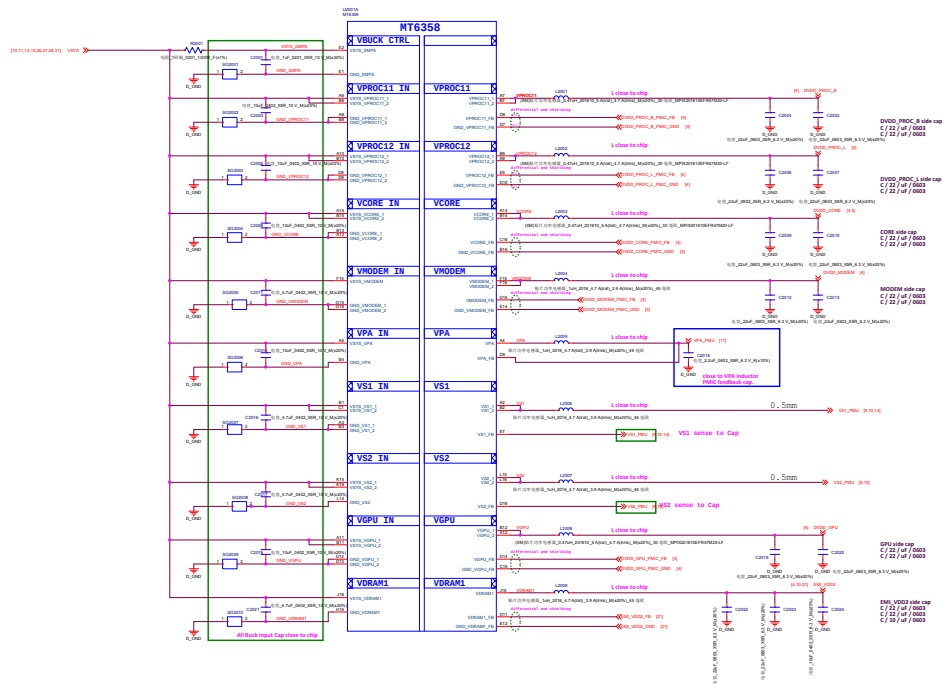
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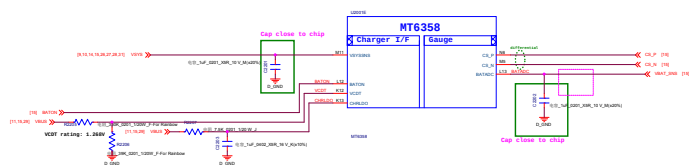


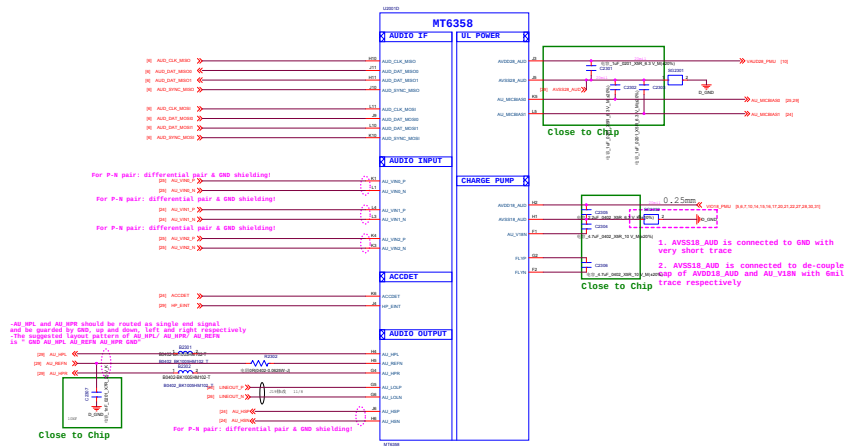
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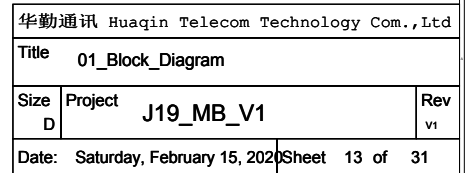


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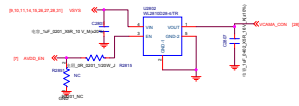




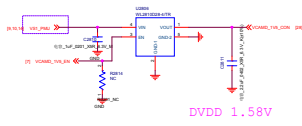
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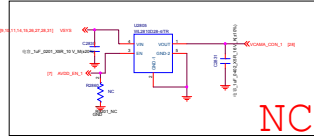
LDO for Dual Camera (VCAMA_EXT) 2.8V



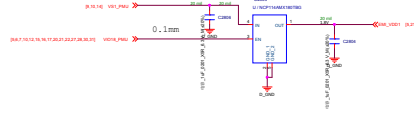
LDO for Depth Camera (VCAMD_EXT1) 1.58V



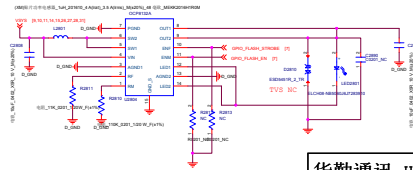
LDO for Dual Camera (VCAMA_EXT2) 2.8V



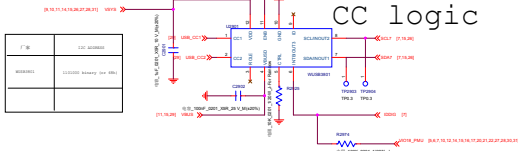
Ext. EMI_VDD1 for LPDDR4X VDD1



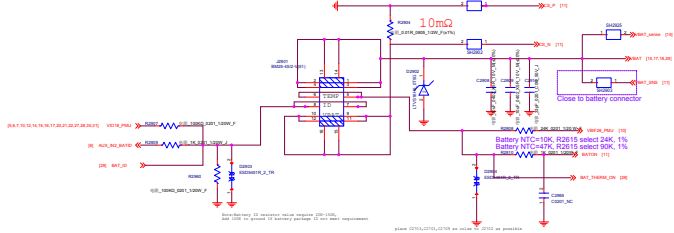
Flash LED 5V Boost



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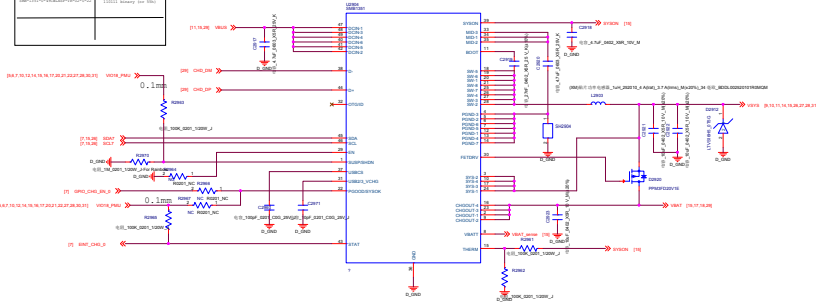
BATTERY CONNECTOR



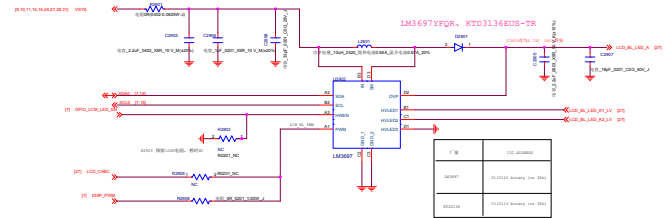
Schematic design notice of "29_EXT_WLED+DB+FL+CHG+PD" page.

Note 29-1: For better ESD or surge performance we need choose suitable device for system protection. Please refer to [Surge device selection guide V2.0] provide by MTK.

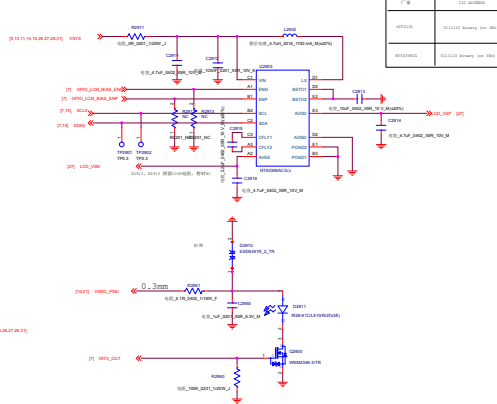
SW Charger



LCM Backlight LED Driver



LCM Bias



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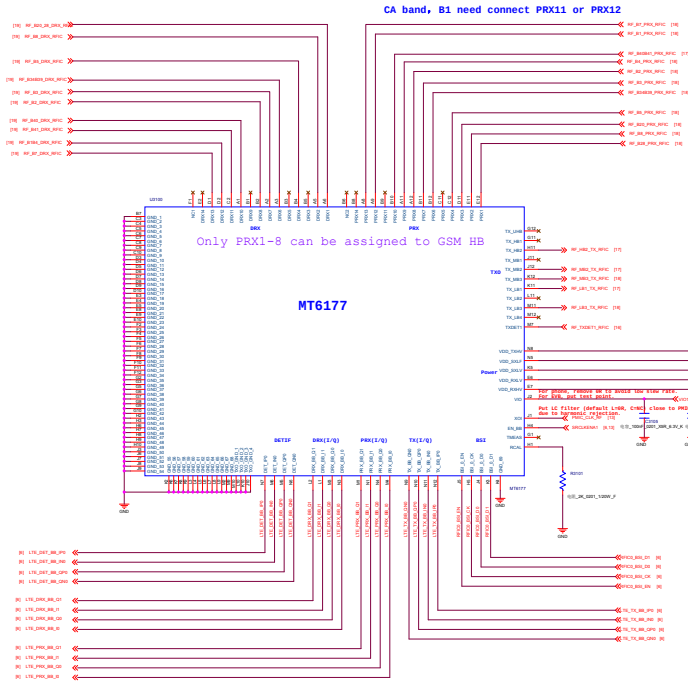
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Attenuator for Detector

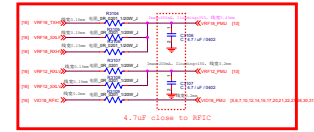
Transceiver



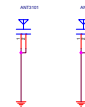
The linear detection range is 0-20dBm, coupling factor is 10-20dB.



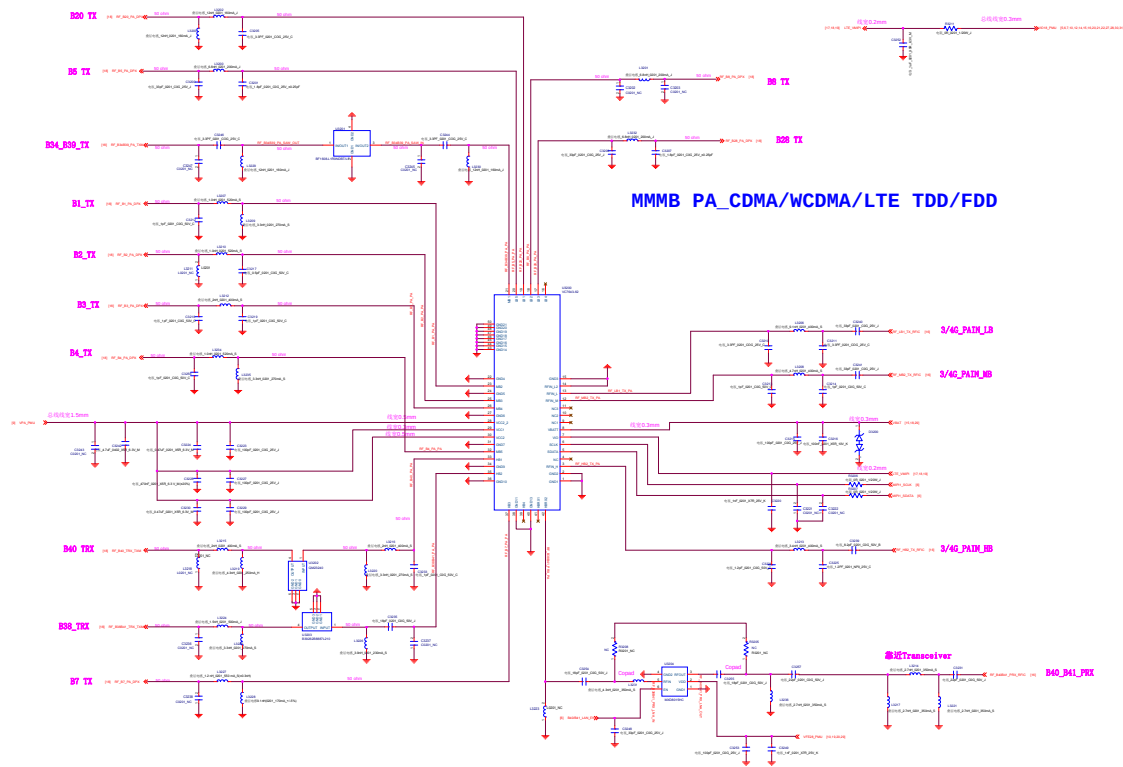
Power For MT6177



GND

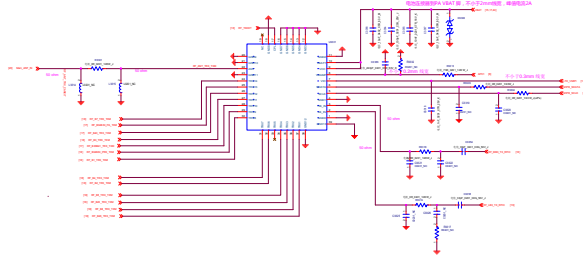


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Title 31_RF_MT6177_Pin_Out		
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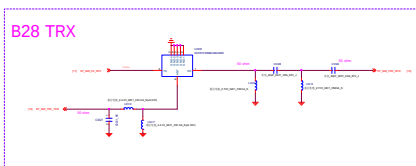
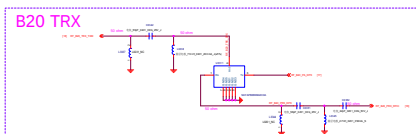
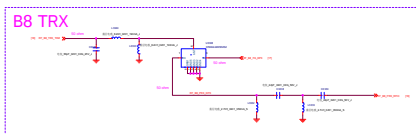
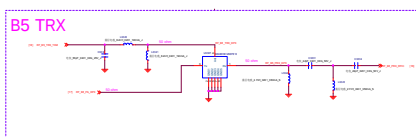


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Title 32_RF_MT6177_RF_TX		
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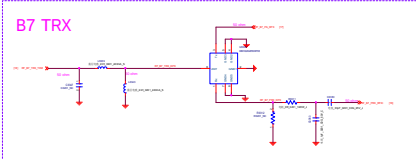
MAIN_ANT



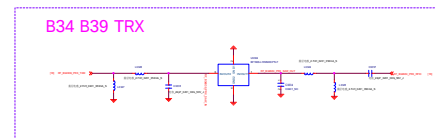
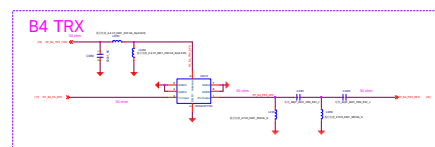
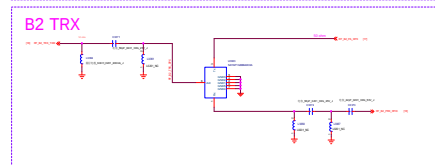
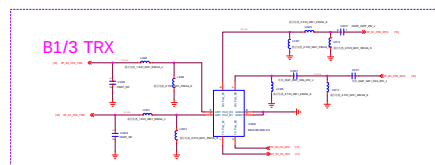
TRX_LB



TRX_HB

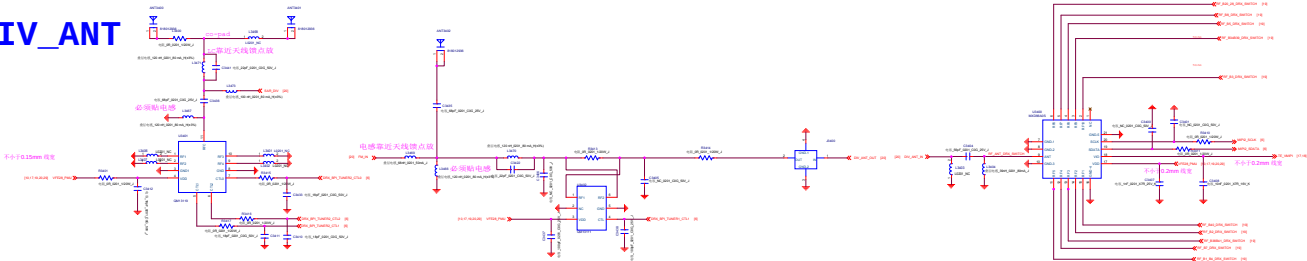


TRX_MB

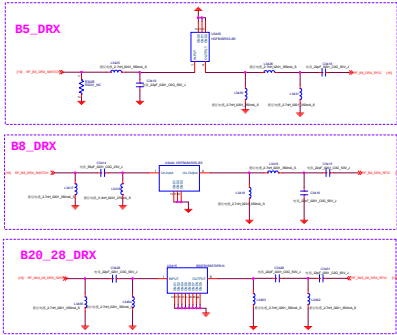


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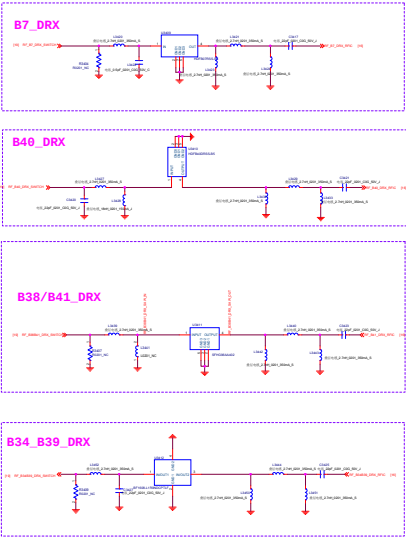
DIV_ANT



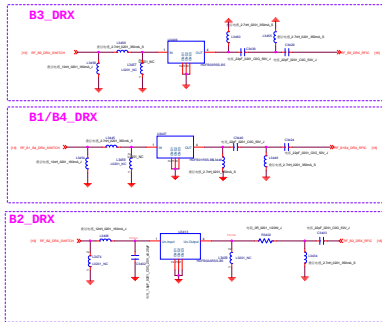
DIV_LB



DIV_HB



DIV_MB



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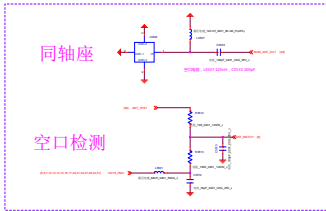
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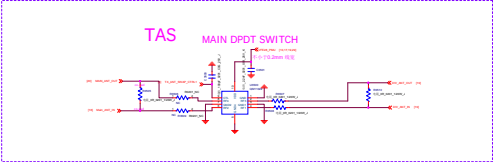
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1

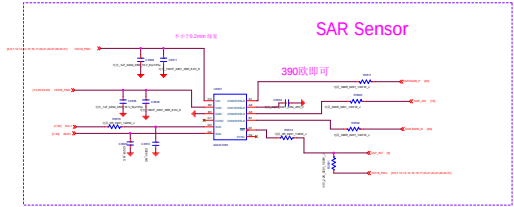
空口检测



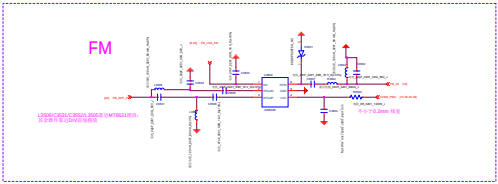
DPDT



SAR Sensor



FM



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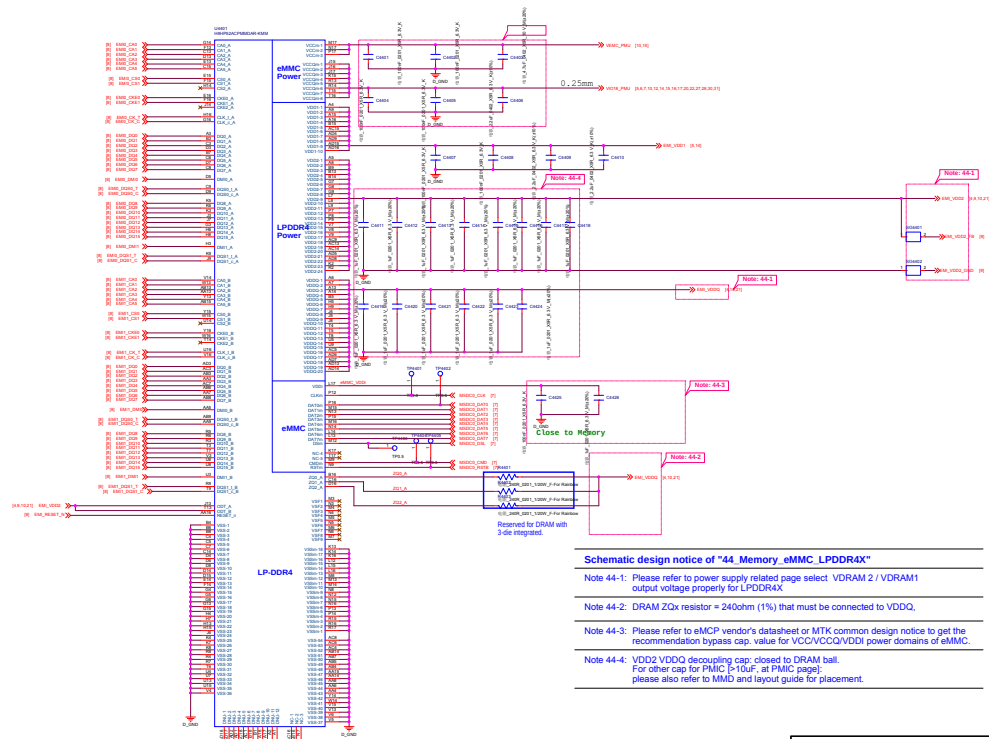
5

4

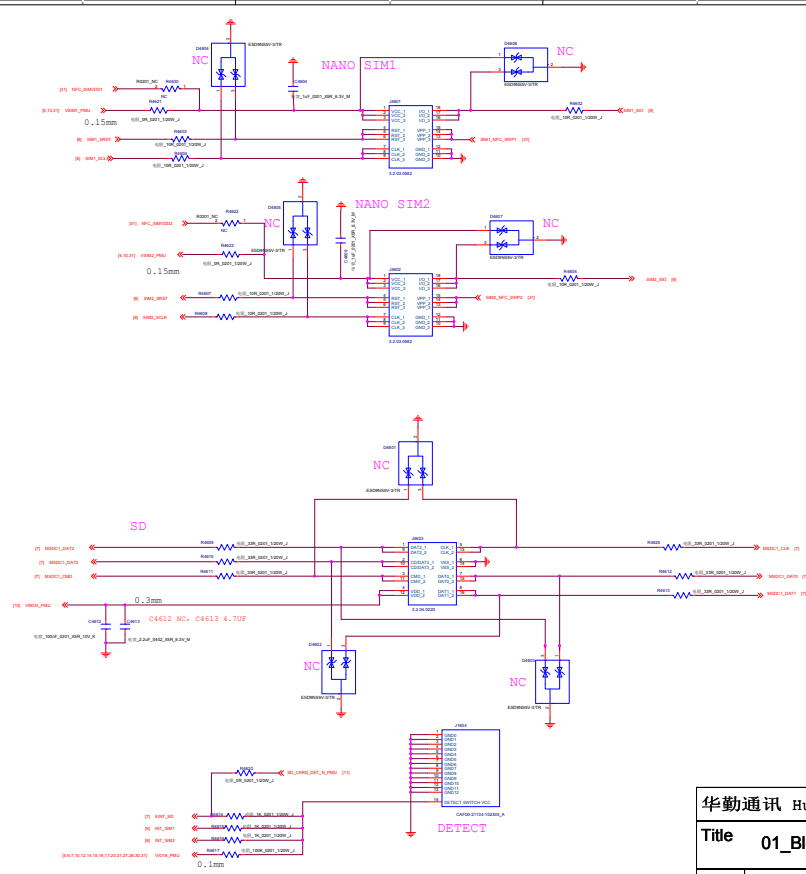
3

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1



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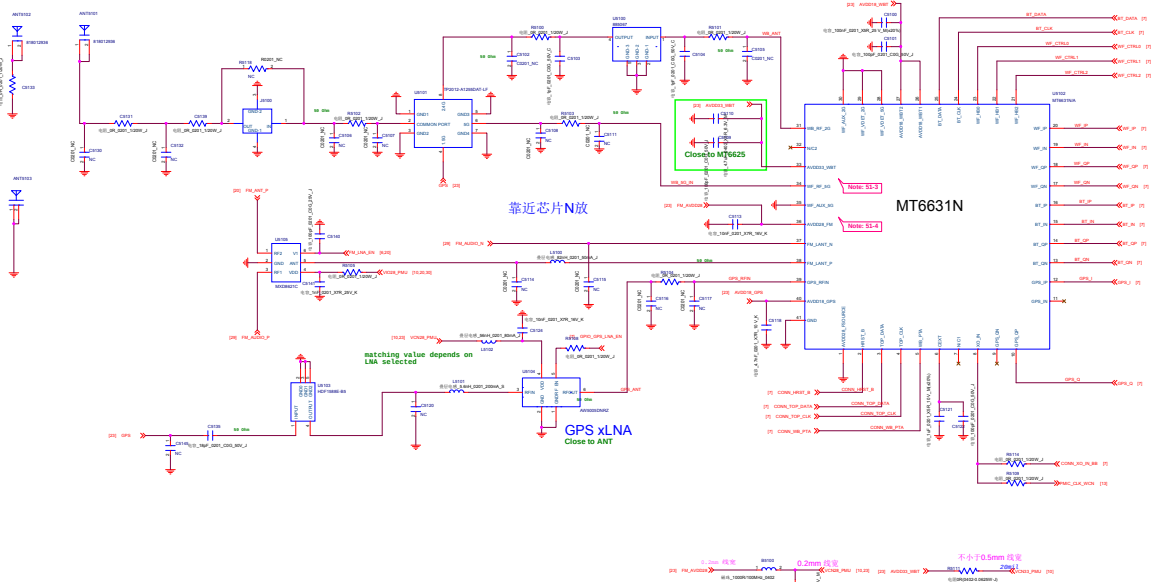


Note 46-1:

[Schematic design notice of "46_MEMORY_SD Card" page.](#)

For better ESD performance, please select suitable components for system protection.

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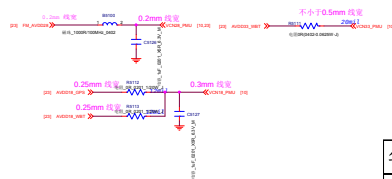
Schematic design notice of "51_CONNECTIVITY_CONSYS_MT6631"

Note 51-1: For RS015 size, please select 0402 size or larger one

Note 51-2: Please refer to MT6785 Baseband design notice for VCN33 LDO selection guide

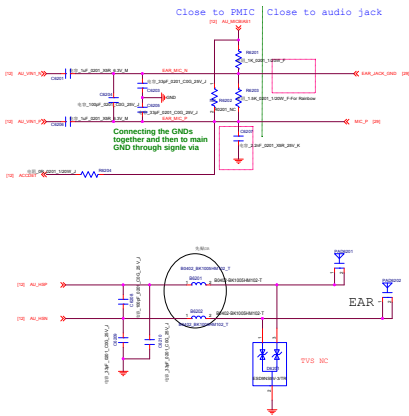
Note 51-3: If WFI 5G not support, connect pin 34(WF_RF_SG) to GND

Note 51-4: Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support



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Earpone Microphone



Schematic design notice of "62_PERI_AUDIO_IO" page.

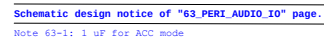
Note 62-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "SLM188D1032N1" for high THD performance (-95dB) but this BOM change will results in FM RSSI 10dB degraded .

Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

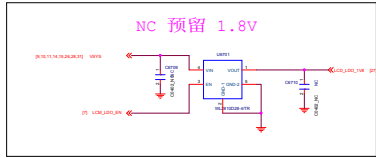
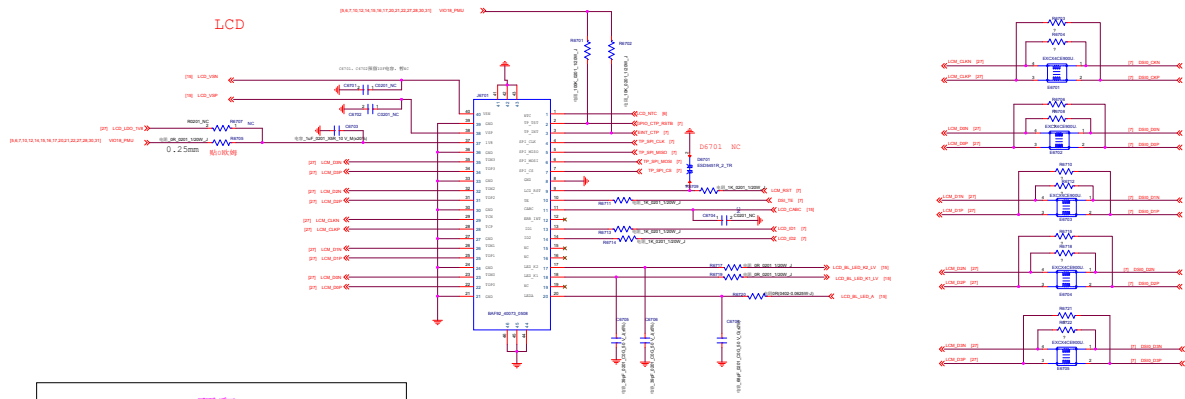
Note 62-3:

Part #	Value	Value
62-1	100nF	100nF
62-2	100nF	100nF

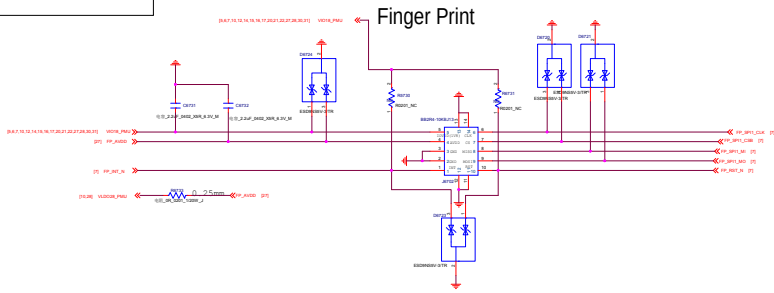
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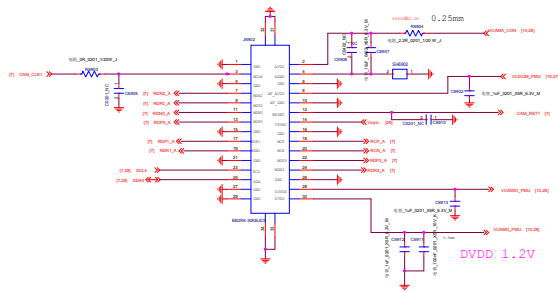
CAP tolerance above 50V!!!



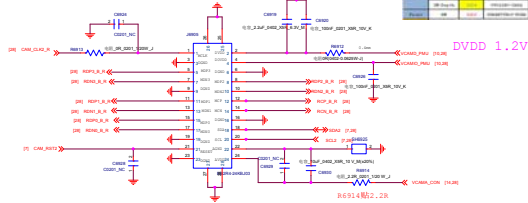
LCD-BACKLIGHT

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Main MONO Camera 13M



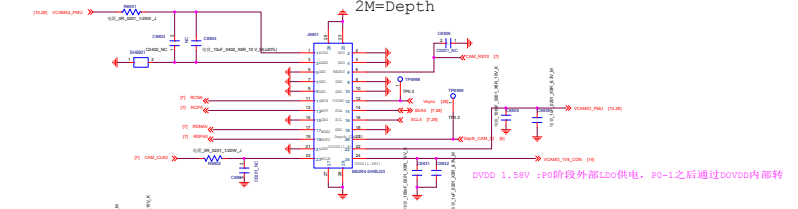
Front Camera - 8M



贴R6967、R6968，其他器件NC

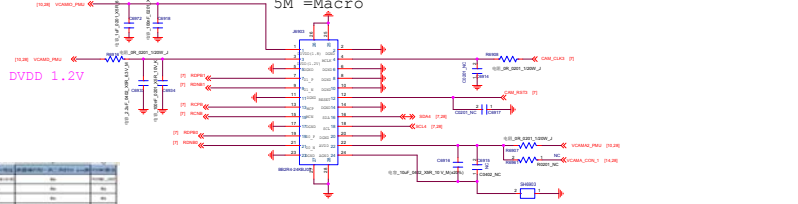
Pin	Function	Value
1	VDD	1.2V
2	VDDA	1.2V
3	VDDM	1.2V

2M=Depth

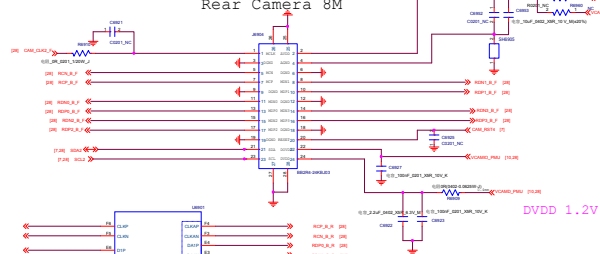


DVDD 1.58V : P0阶段外部LDO供电，P0-1之后通过DVDD内部转

5M=Macro



Rear Camera 8M



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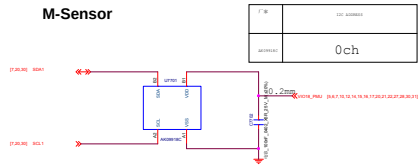
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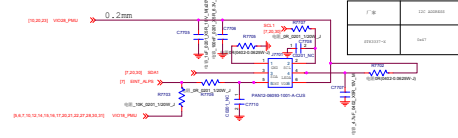
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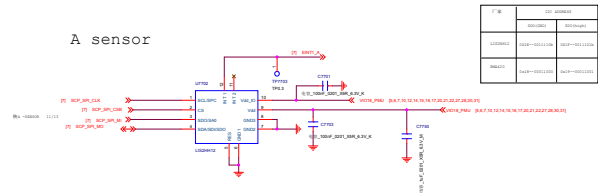
M-Sensor



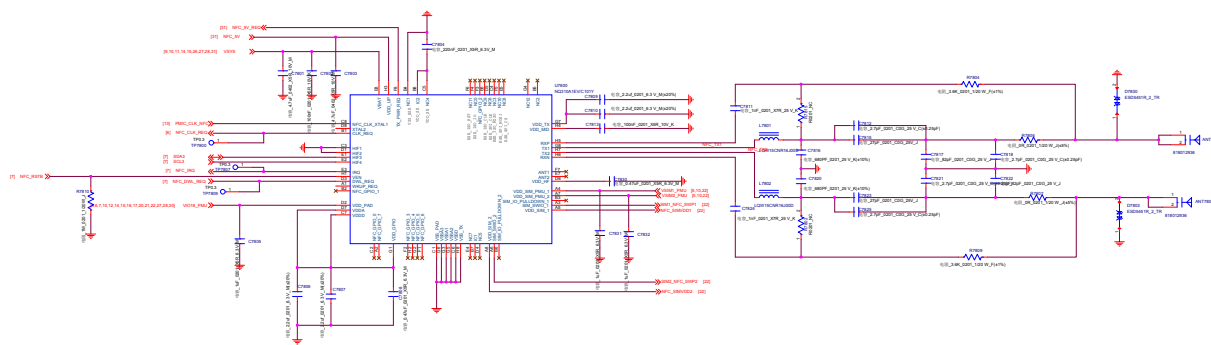
ALS PS STK3337-X



A sensor

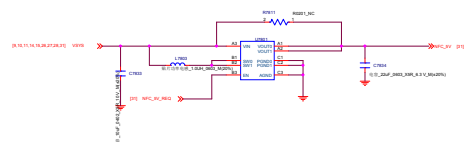


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NFC DCDC

DCDC no part number



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